

Announcements

- ▶ Any questions/issues?

- ▶ Schedule:

Today (Follow-up): Question 2 from “Forcing and Timescales” discussion — Exponential Growth and Doubling

Today (New): Modeling and forecast of climate change

Tomorrow (Lab): Computational Modeling (web app)

Thursday (Discussion): More Computational Modeling

Next Week: Oceans and Ice

Today's Reading Quiz (3 min)

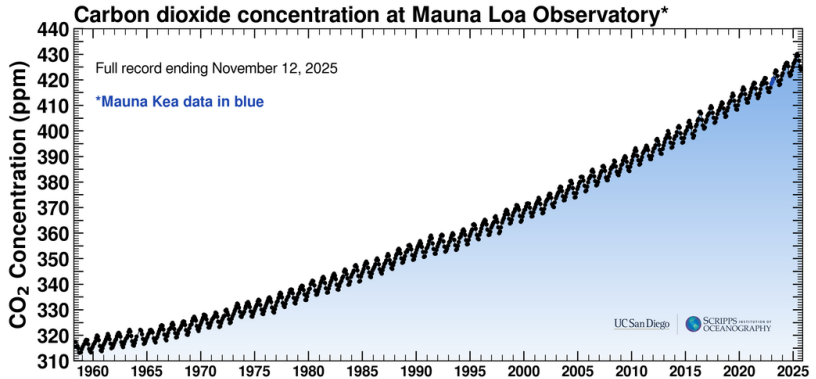
No reading for today, but please prepare for tomorrow and Thursday by:

- ▶ Reading Chapter 8 from Dessler (“Predictions of Future Climate Change”)
- ▶ Watching the [tutorial video](#) for the climate simulator **EN-ROADS** (find link on “Weekly Reading Assignments” page)

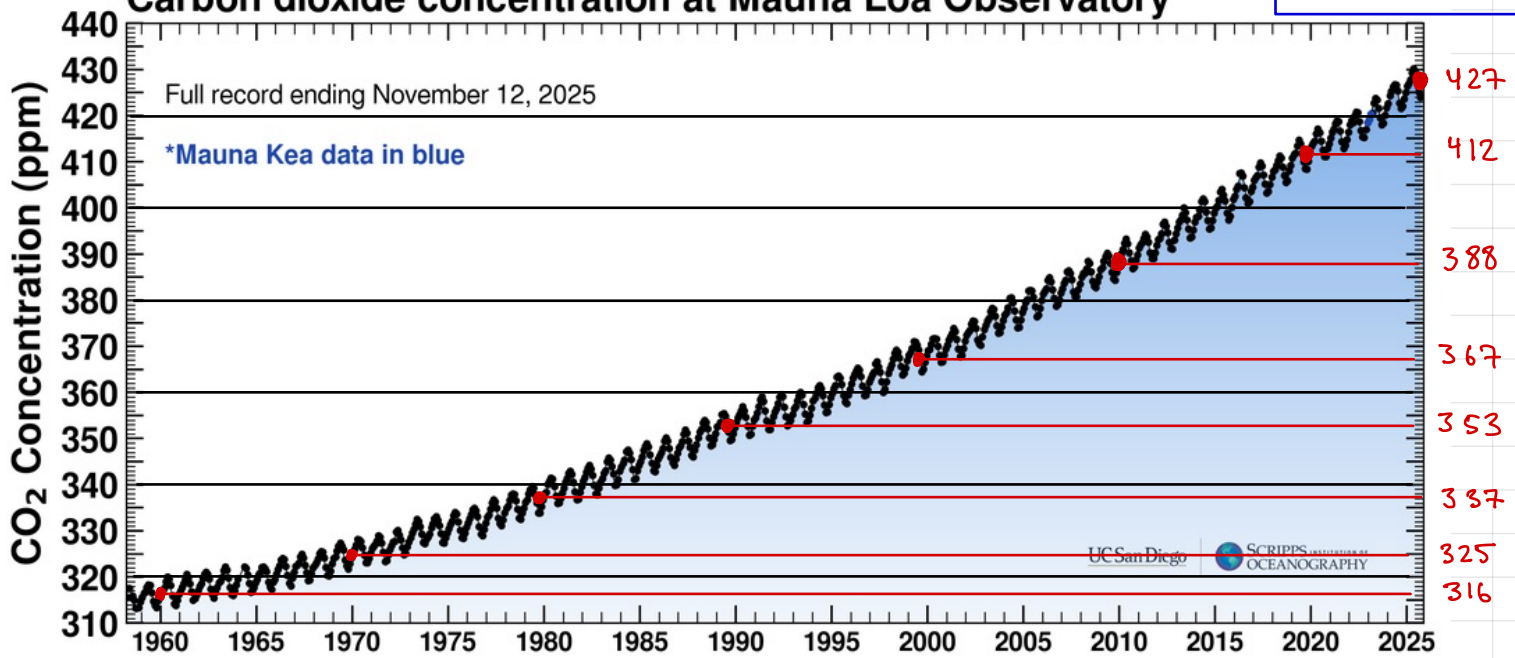
Final Poster Project

- ▶ Choose a topic that is interesting to you and make a poster to present (5min) to the class during our final exam period (Thurs, Dec 11, 10am).
- ▶ Work on your own or with a partner.
- ▶ Think of it as a Discussion/Lab/Lecture that you create, explaining some topic to yourself and us.
- ▶ **Talk to me by Thanksgiving (Tues Nov 25)** to verify your project topic (**worth one Discussion assignment**).
- ▶ In the last week of class you will have time (lecture and discussion) to work on your final poster project.

Exponential Growth of CO₂ concentration



Carbon dioxide concentration at Mauna Loa Observatory*



DATE	C_{CO_2}	$\log_{10}[C_{CO_2}]$
1960	316	2.500
1970	325	2.512
1980	337	2.528
1990	353	2.548
2000	367	2.565
2010	388	2.589
2020	412	2.615
2025	427	2.630

WHAT DOES " $\log_{10}(\text{CONC}_{CO_2})$ " MEAN?

$$\log_{10}(X) = Y \Leftrightarrow 10^Y = X$$

EG. $10^2 < 10^{2.5} < 10^3$ $\rightarrow$ 325

(100) (1000)

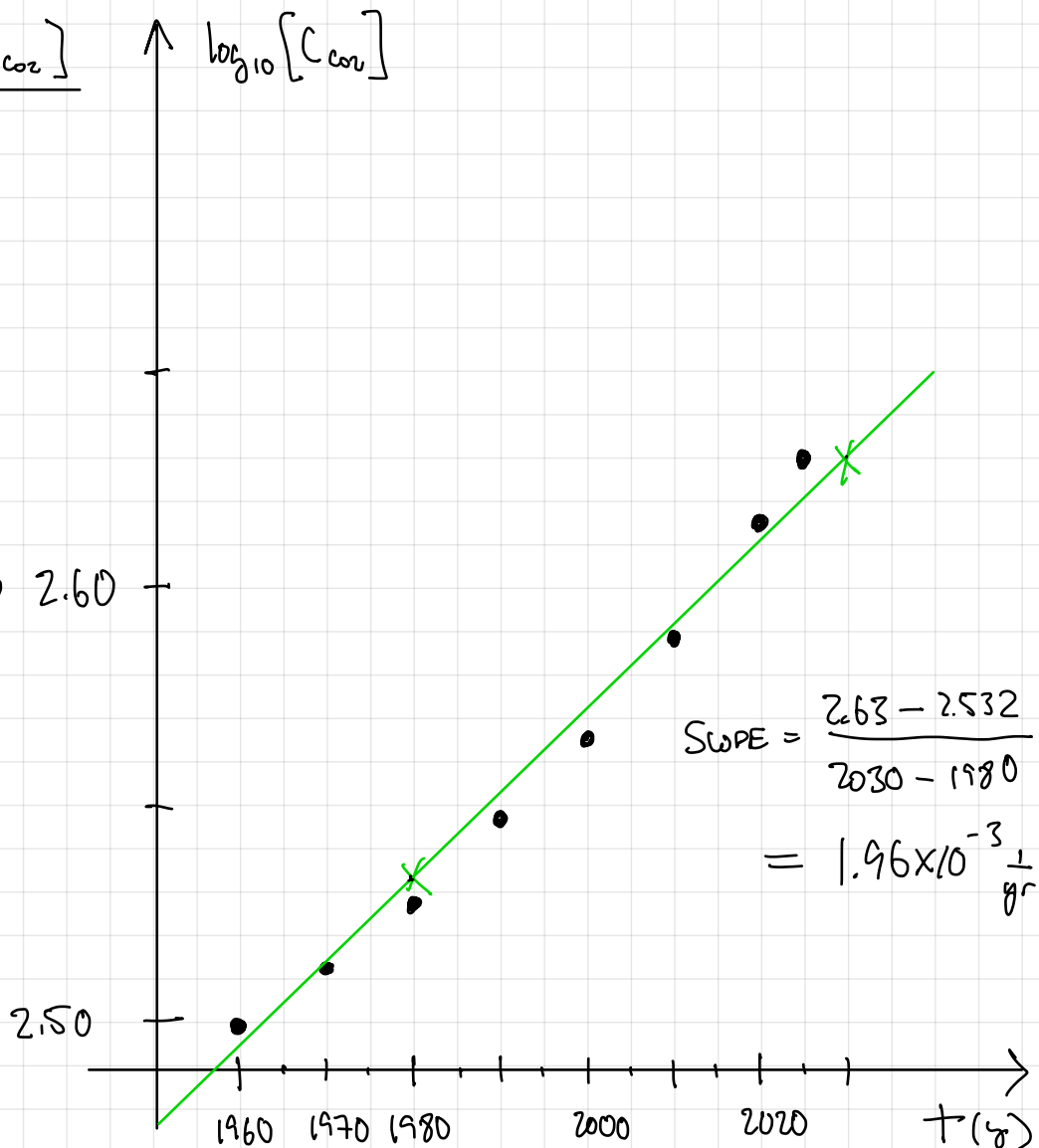
HOW DO WE VERIFY AN
EXPONENTIAL?

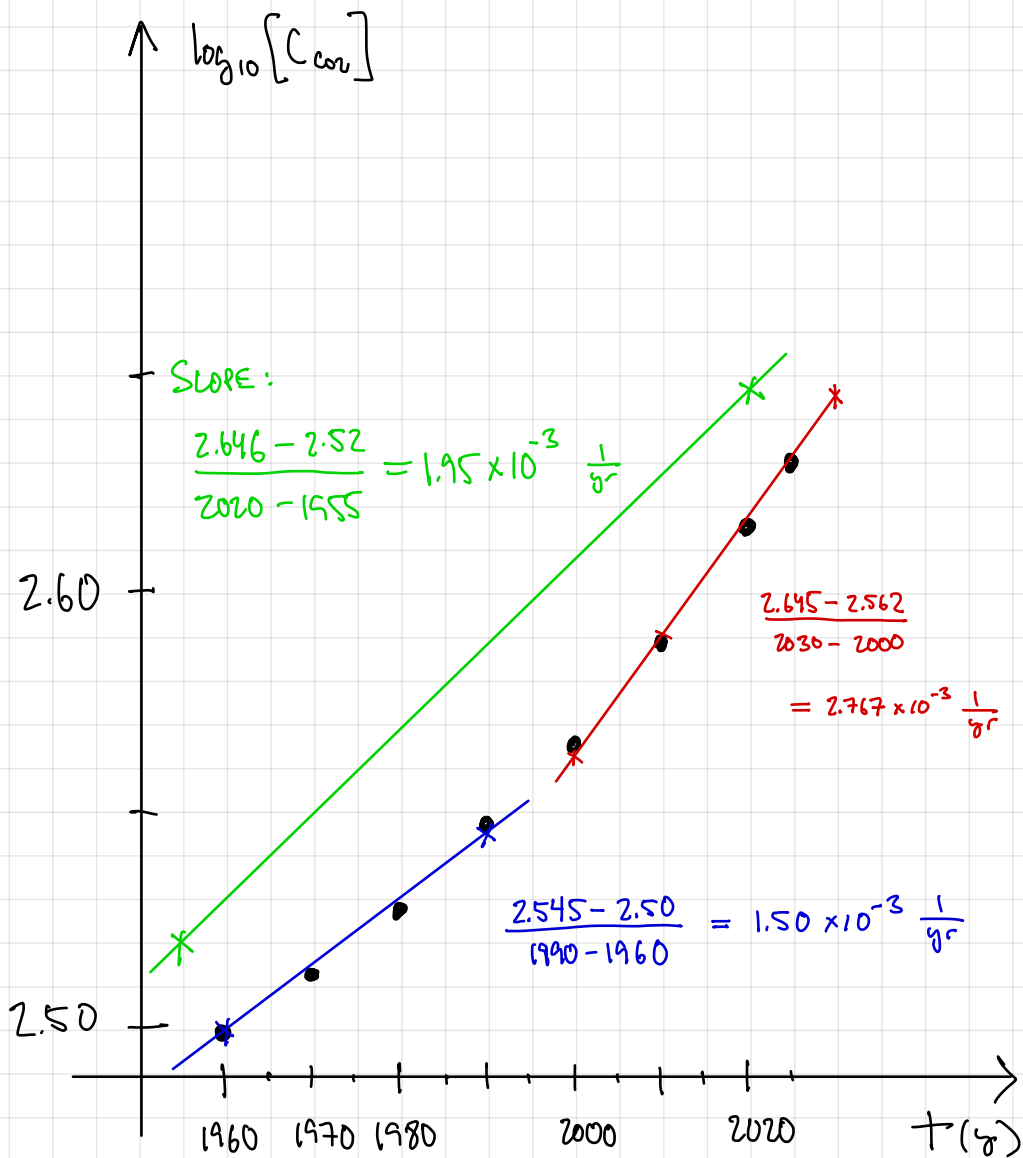
IF WE PLOT THE DATA
 C vs. t

As

$\log_{10}[C]$ vs. t

THEN A STRAIGHT LINE IS AN EXPONENTIAL





SO THE SLOPES ABOVE GIVE:

$$t_{\text{DOUBLE}}(1960-2025) = \frac{\log_{10}(2)}{1.95 \times 10^{-3} \frac{1}{\text{yr}}} = \underline{154 \text{ yr}}$$

$$t_{\text{DOUBLE}}(1960-1990) = \frac{\log_{10}(2)}{1.5 \times 10^{-3} \frac{1}{\text{yr}}} = \underline{200 \text{ yr}}$$

$$t_{\text{DOUBLE}}(2000-2025) = \frac{\log_{10}(2)}{2.77 \times 10^{-3} \frac{1}{\text{yr}}} = \underline{110 \text{ yr}}$$

DOUBLING TIME

$$C(t) = C(t=0) \cdot 10^{s \cdot t}$$



$$\log C(t) = \log[C(0)] + s \cdot t$$

$$(Y = b + mX)$$



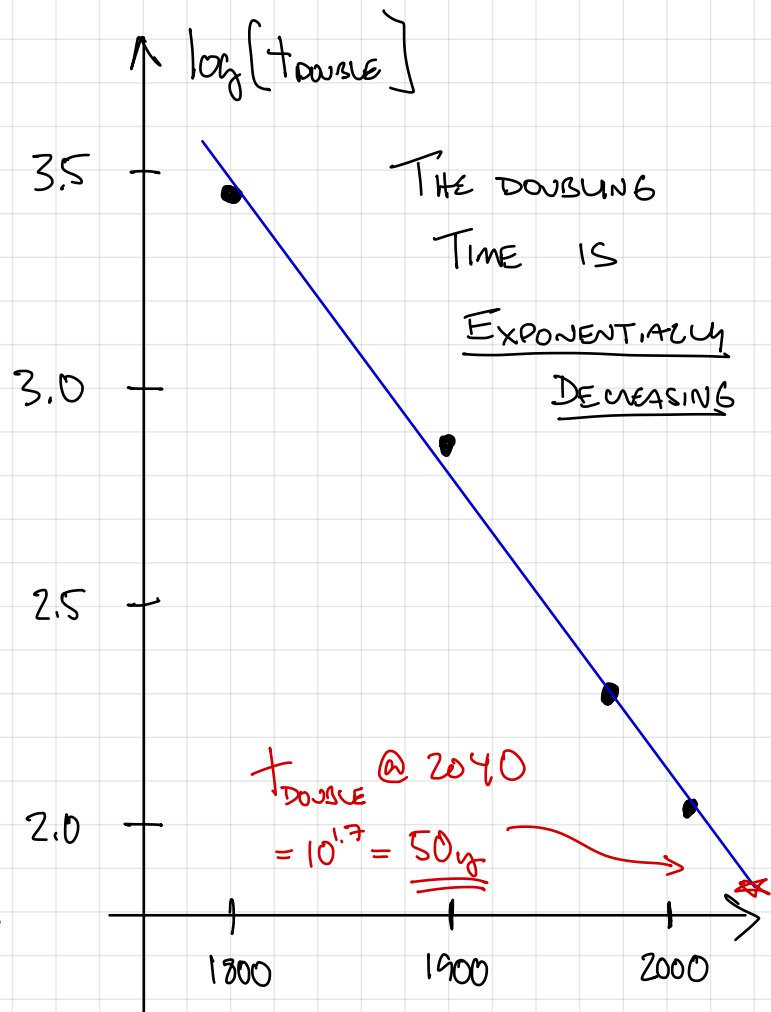
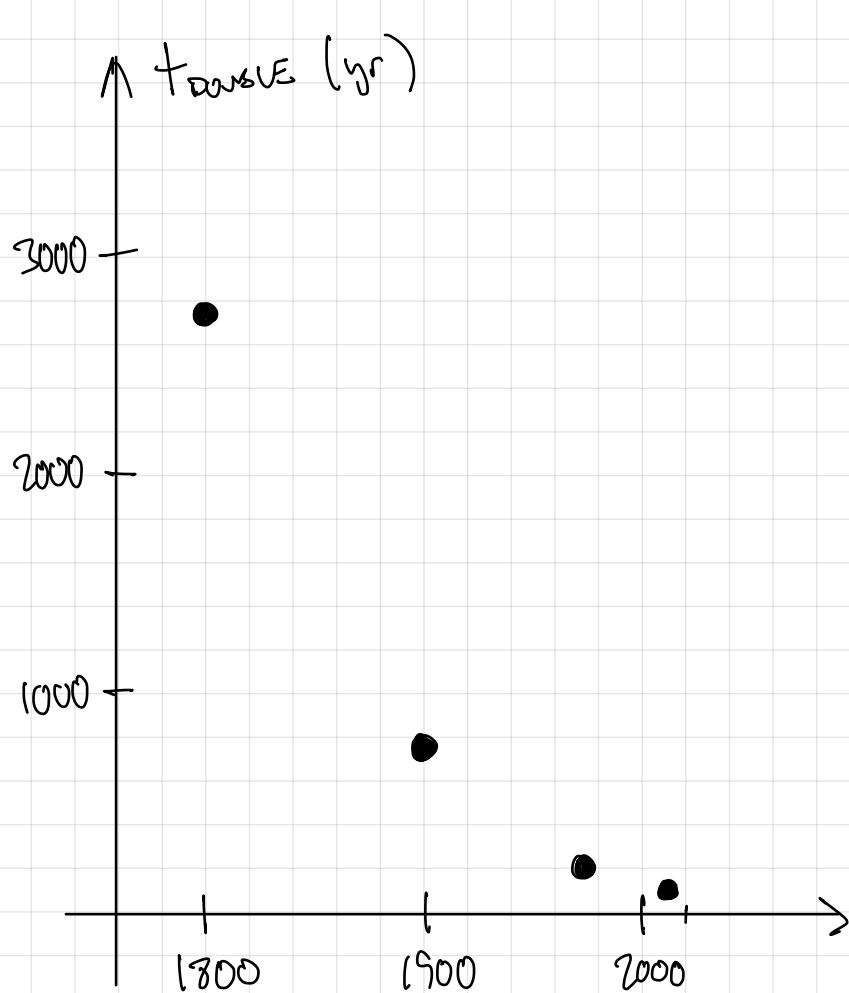
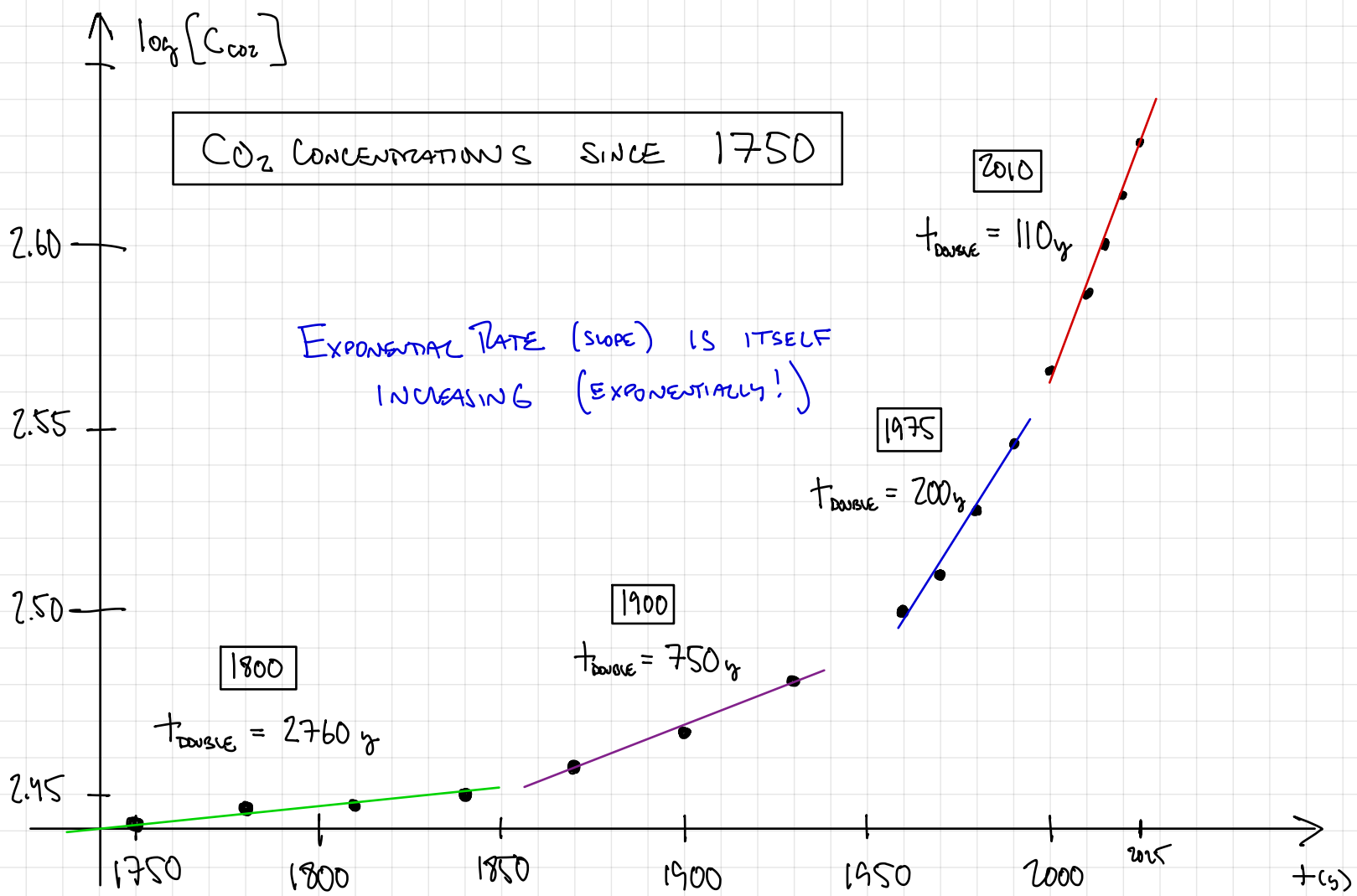
$$C(t) = C(0) 10^{s \cdot t}$$

$$2 \times C(0) = C(0) 10^{s \cdot t}$$

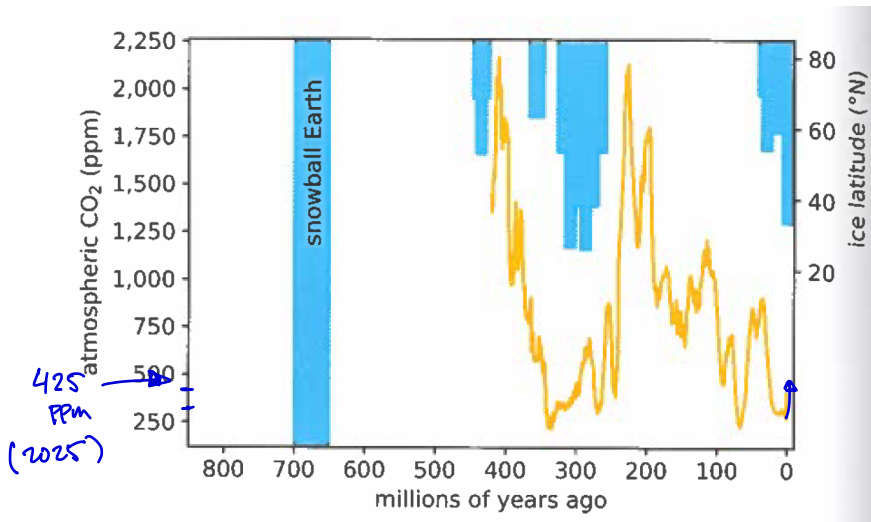
$$2 = 10^{s \cdot t}$$

$$\log_{10} 2 = s \cdot t$$

$$t_{\text{DOUBLE}} = \frac{\log_{10} 2}{s}$$



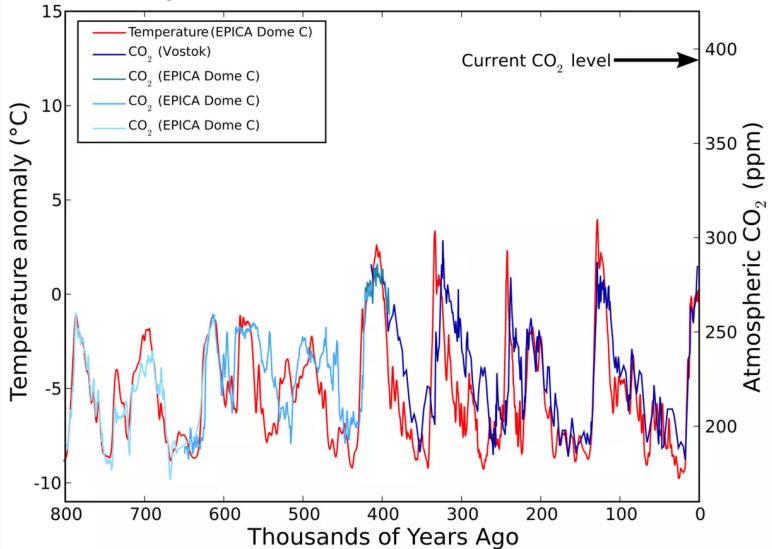
CO₂ concentration on planetary timescales



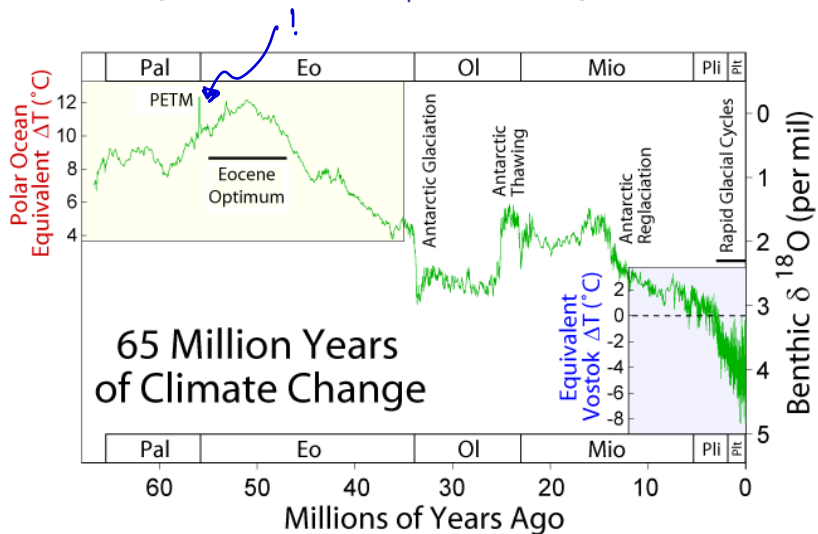
Dessler Fig 7.7

CO₂ and temperature

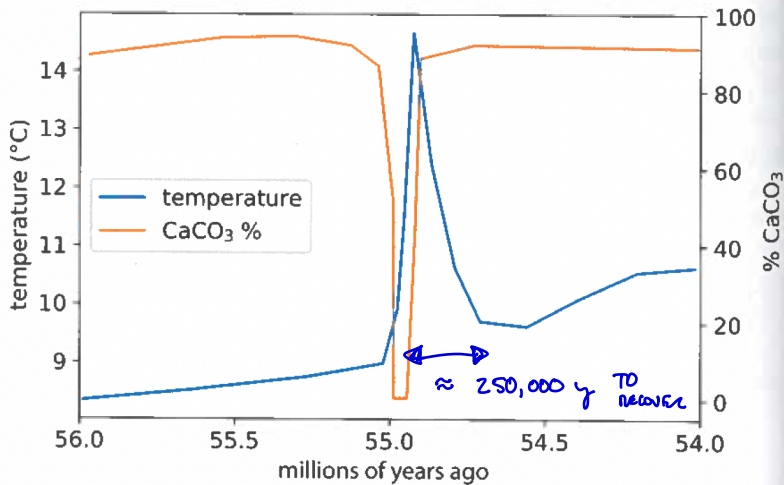
Temperature and CO₂ Records



The last big temperature/ CO_2 surge: PETM

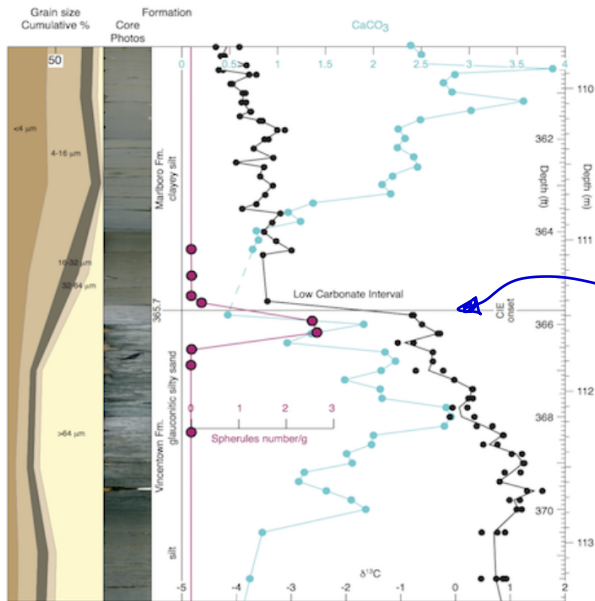


PETM timescale



Dessler Fig 7.8

PETM timescale



ONSET OF
PETM
WAS POSSIBLY
< 2,000 y

(CLOSEST ANALOG TO
OUR CONTEMPORARY
TEMPERATURE CHANGE
... BUT LESS DRAMATIC)

Timescales of contemporary climate change

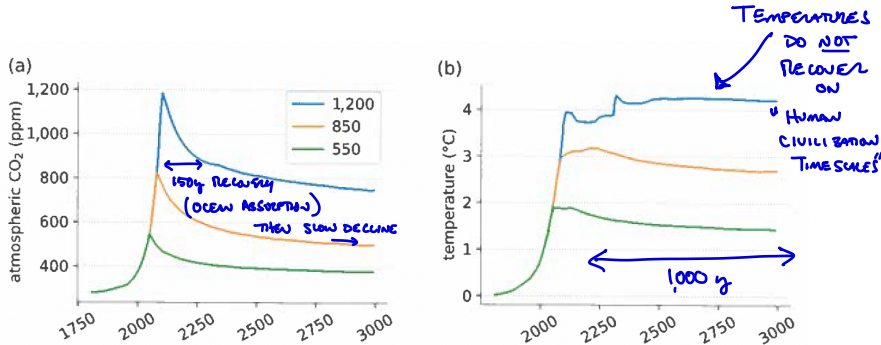


Figure 8.5 (a) Amount of carbon dioxide in the atmosphere as a function of time, for the next 1,000 years. Carbon dioxide emissions rise at 2 percent/year until it hits a peak abundance (550, 850, and 1,200 ppm); then emissions are decreased instantly to zero. (b) The temperature time series corresponding to each carbon dioxide time series. Adapted from Figure 1 of Solomon et al. (2009).

Dessler

INCREASING CO₂ EMISSIONS
THEN HALT TO ZERO EMISSIONS.